Reproducing and Evaluating Uncertainty-Aware Neural Networks for Train Delay Prediction: An Empirical Study

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***Abstract*—Train delay prediction is an important problem in railway operations because delays affect passenger journeys, resource allocation, and the reliability of railway services. Although machine learning models can estimate train delays, a point prediction alone does not indicate how reliable the prediction is. Uncertainty-aware prediction therefore provides additional information that may support operational decision-making.**

This paper presents an empirical implementation and evaluation of an uncertainty-aware neural network approach for train delay prediction, motivated by the framework proposed by Spanninger et al. The implementation uses the train-delay prediction repository released by the original research group and evaluates the model against simple statistical baselines. The reported dataset contains 60,000 observations and 46 columns, with `final_delay` used as the prediction target. The evaluated neural-network configuration contains two hidden layers of 128 units, a learning rate of 10^{-5}, and a dropout rate of 0.5.

In the current experiment, the uncertainty-aware neural network achieves a mean absolute error (MAE) of approximately 24.045 and a mean squared error (MSE) of approximately 1001.015. The reported simple mean and median baselines produce an MAE of approximately 85.314, while the simple-average baseline produces an MAE of approximately 85.538. The neural-network result is also examined using prediction-interval coverage, interval width, and likelihood-related measures. These findings are preliminary and are not presented as an exact reproduction of the original paper, because the dataset provenance, experimental equivalence, and metric implementation require further verification. The study establishes an initial experimental baseline for more systematic evaluation of uncertainty-aware railway delay prediction.

***Index Terms*—Train delay prediction, railway operations, uncertainty estimation, neural networks, dropout, predictive intervals, machine learning, empirical evaluation.**

I. Introduction

Railway transportation systems operate under interconnected constraints involving train schedules, infrastructure availability, route conflicts, weather conditions, and operational disturbances. A delay affecting one train may influence subsequent services, making delay prediction a challenging problem in railway planning and control.

Research on train delay prediction has explored statistical models, machine learning, deep learning, and methods that explicitly represent operational dependencies. Existing work has investigated delay propagation, route conflicts, weather information, operational sequences, and disturbance-related factors [1], [2], [4], [6], [8]–[11]. Reviews of data-driven and train-delay prediction methods further demonstrate the diversity of modelling strategies and evaluation settings [35], [36].

Despite progress in prediction accuracy, an important distinction remains between predicting a delay and estimating the reliability of that prediction. A model may produce a numerically accurate estimate for one observation while remaining uncertain about another. In operational settings, this distinction matters because decision-makers may need to identify predictions that are sufficiently reliable for planning and predictions that warrant additional attention.

Uncertainty-aware neural networks address this issue by combining point predictions with information about predictive uncertainty. The work of Spanninger et al. investigates uncertainty-aware neural networks for train delay prediction and examines how the predictability of train delays changes dynamically [3]. This work provides the principal research motivation for the present study.

A. Problem Statement

The primary objective of this paper is to implement and evaluate an uncertainty-aware neural-network approach for train delay prediction and compare its observed performance with simple statistical baselines.

The study focuses on three questions:

1) How does the evaluated neural network perform in terms of point-prediction error compared with simple statistical baselines?
2) What do the reported uncertainty measures indicate about prediction intervals and the reliability of the evaluated model?
3) What implementation, data, and evaluation checks are necessary before the experiment can be considered a faithful reproduction of the original research?

B. Objectives

The objectives of the study are:

- To examine and execute the publicly available implementation associated with uncertainty-aware train delay prediction.
- To evaluate the selected neural-network configuration using point-prediction and uncertainty-related metrics.
- To compare the observed results with simple statistical baselines.
- To document the experimental configuration, preliminary results, and limitations that affect reproducibility.
- To establish a foundation for repeated-run evaluation and further investigation of uncertainty calibration and prediction reliability.

C. Contributions

The contributions of this paper are as follows:

1) An empirical implementation and initial evaluation of an uncertainty-aware neural-network approach for train delay prediction.
2) A comparison of the currently reported prediction errors against simple statistical baselines.
3) An examination of prediction-interval coverage, interval width, and likelihood-related evaluation measures alongside point-prediction error.
4) A discussion of the limitations associated with dataset provenance, metric interpretation, and reproducibility.

The current results should be interpreted as an initial experimental evaluation. A definitive reproduction claim requires confirmation that the data, preprocessing, evaluation protocol, and metric definitions match those used in the original study.

II. Related Work

A. Train Delay Prediction

Train delay prediction has been investigated using statistical forecasting, machine learning, and deep learning. Research has considered the propagation of delays across connected services, the influence of route conflicts, and the relationship between train operations and environmental conditions [1], [2], [6]–[11].

Hybrid deep learning models have been proposed to represent complex patterns in train-delay data [2]. Other approaches investigate the use of graph attention networks to explain delay propagation and model interactions between railway operations [4]. Transformer-based approaches have also been explored for parallel prediction of delay propagation [5]. These methods illustrate the importance of temporal dependencies and interactions between trains.

Further research has investigated train interactions and operational dependencies [29], as well as cause-specific primary delays in high-speed railway operations [30]. Such studies are relevant because delay prediction performance depends not only on the learning algorithm but also on the operational variables and relationships represented in the data.

Reviews of train-delay prediction methods provide a broader overview of the available modelling strategies and their limitations [35], [36]. Together, these studies establish train-delay prediction as a data-driven problem in which prediction accuracy, operational context, and evaluation methodology must be considered together.

B. Uncertainty-Aware Prediction

Conventional regression models generally produce a point estimate of the target variable. Although point estimates are useful for forecasting, they do not directly describe the uncertainty associated with individual predictions.

Spanninger et al. investigated uncertainty-aware neural networks for train delay prediction and examined the dynamic predictability of delays [3]. Their work motivates evaluating uncertainty measures in addition to traditional regression metrics.

The broader machine-learning literature has also explored deep-learning architectures that represent complex relationships in sequential and structured data. Graph convolutional networks provide a mechanism for learning from graph-structured relationships [31], while the Transformer architecture uses attention mechanisms for sequence modelling [32]. These approaches provide relevant methodological context, although they are not treated as direct experimental baselines in the present implementation.

C. Vision-Based Railway Inspection

Computer vision is another important research direction in railway monitoring. Existing studies investigate railway track defects, surface damage, fastener defects, track components, and foreign-object detection [12]–[23].

YOLO-based methods have been adapted for rail surface defect detection, foreign-object detection, and railway obstacle identification [13], [14], [17], [18]. Other studies examine supervised and self-supervised defect detection, pseudo-labels, synthetic data, and methods designed to address class imbalance [12], [15], [20], [21]. Lightweight detection architectures and feature-fusion techniques have also been investigated for railway inspection [22], [23], [28].

Reviews of rail-track condition monitoring and machine-vision-based inspection summarize the development of these methods and their practical challenges [24]–[27]. The general object-detection approach introduced by YOLO and residual learning architectures provide additional foundational context [33], [34].

These studies are relevant to the broader development of intelligent railway monitoring systems. However, visual defect detection and train delay prediction are distinct tasks. A detected defect does not, by itself, establish that a particular train will experience a measurable delay. Such a relationship requires appropriately linked infrastructure and operational data, which is outside the verified scope of the present experiments.

D. Research Gap and Motivation

The literature demonstrates substantial work on train delay prediction, railway operational dependencies, uncertainty

estimation, and vision-based railway inspection. However, prediction error alone does not establish whether an uncertainty-aware model produces useful and reliable uncertainty estimates.

This paper therefore concentrates on the empirical evaluation of an uncertainty-aware neural network and simple statistical baselines. Rather than claiming a new prediction architecture, the study emphasizes transparent reporting of the current implementation, observed metrics, and outstanding reproducibility checks.

III. Methodology

A. Research Approach

The study follows an implementation-and-evaluation approach. The publicly available train-delay prediction repository associated with the base paper is used as the starting point [3], [37]. The selected model configuration is executed, and the resulting performance is examined using the available evaluation outputs.

The methodology consists of the following stages:

1) Inspect the implementation and available data.
2) Execute the selected model configuration.
3) Record the point-prediction and uncertainty metrics.
4) Compare the neural-network results with simple statistical baselines.
5) Identify the checks required for a reproducible and scientifically defensible evaluation.

The current experiment is not assumed to be identical to the original authors' experiment. Such equivalence depends on verifying the data source, preprocessing pipeline, model configuration, train-validation-test splits, and metric calculations.

B. Dataset Description

The current local dataset, reported as `data_enriched.csv`, contains 60,000 observations and 46 columns. The target variable used for the prediction experiment is `final_delay`. The reported feature matrix contains 37 input features.

The currently recorded split sizes are shown in Table I.

TABLE I
Reported Dataset Configuration

Property	Reported Value
Dataset file	`data_enriched.csv`
Total observations	60,000
Total columns	46
Input features	37
Target variable	`final_delay`
Training observations	44,980
Validation observations	5,984
Test observations	9,036
Missing values	None reported

The split sizes sum to 60,000 observations. However, the original research repository notes that the dataset used in the published study is not publicly available in its original form. Consequently, the provenance of the local dataset and its equivalence to the original experimental data must be verified before claiming an exact reproduction.

The statement that no missing values were observed refers to the current inspection of the local dataset. It does not establish that the data are free from all quality issues, leakage, duplicate records, or temporal dependencies across splits.

C. Neural-Network Configuration

The evaluated neural-network configuration is identified in the local implementation as:

`nn-128-128-2-1e-05-0.5`

The recorded configuration is summarized in Table II.

TABLE II
Reported Neural-Network Configuration

Parameter	Value
Model type	Neural network
Hidden-layer widths	128, 128
Number of hidden layers	2
Learning rate	1×10^{-5}
Dropout rate	0.5
Prediction target	`final_delay`

The configuration name and values are taken from the current local execution context. Before final submission, the architecture and inference procedure should be confirmed against the implementation files. In particular, the exact way dropout is used during inference must be verified before describing the uncertainty estimates as Monte Carlo dropout estimates.

D. Baseline Models

Simple statistical baselines are included to provide a reference for interpreting the neural-network results. The currently available outputs include simple mean, simple median, and simple-average baselines.

These baselines are useful as initial reference points, but they are not sufficient to establish superiority over all relevant train-delay prediction methods. A stronger evaluation should also compare against appropriate regression models and should ensure that all baselines use the same target definition, test set, and evaluation protocol.

E. Evaluation Metrics

The experiment records point-prediction metrics and uncertainty-related measures.

1) Mean Absolute Error: Mean absolute error measures the average absolute difference between the observed and predicted values:

$$\mathrm{MAE} = \frac{1}{N} \sum_{i=1}^{N} |y_i - \hat{y}_i| \,, \tag{1}$$

where y_i is the observed target, $\hat{y}_i$ is the prediction, and N is the number of evaluated observations.

2) Mean Squared Error: Mean squared error is defined as:

$$\mathrm{MSE} = \frac{1}{N}\sum_{i=1}^{N}\left(y_i - \hat{y}_i\right)^2. \tag{2}$$

MSE assigns a larger penalty to larger errors because the residuals are squared.

3) Root Mean Squared Error: Root mean squared error is the square root of MSE:

$$\mathrm{RMSE} = \sqrt{\mathrm{MSE}}. \tag{3}$$

For the reported neural-network MSE of approximately 1001.015, the corresponding RMSE is approximately 31.64 in the target's units, assuming the reported MSE uses the original target scale.

4) Prediction-Interval Coverage: Prediction-interval coverage records the proportion of observations for which the realized target falls inside the reported prediction interval:

$$\mathrm{Coverage} = \frac{1}{N}\sum_{i=1}^{N}\mathbb{I}\left(y_i \in [L_i, U_i]\right), \tag{4}$$

where L_i and U_i denote the lower and upper bounds of the interval and $\mathbb{I}(\cdot)$ is an indicator function.

Coverage must be interpreted together with interval width. A very wide interval may achieve high coverage while providing limited practical information.

5) Mean Prediction-Interval Width: The mean interval width is expressed as:

$$\mathrm{MPIW} = \frac{1}{N}\sum_{i=1}^{N}(U_i - L_i). \tag{5}$$

A narrower interval is not automatically better. Its usefulness depends on whether it achieves an appropriate coverage level and whether the intervals are calibrated.

6) Additional Recorded Measures: The current execution outputs also contain `Likely_15`, `Likely_30`, `Likely_45`, `mean_unc`, a quantile parameter, a best-factor parameter, and correlation statistics.

The precise definitions and units of these measures must be confirmed from the implementation's metric functions before they are given a stronger statistical interpretation. In particular, likelihood-related measures should not be interpreted as calibrated probabilities without verifying their definitions.

IV. Results and Analysis

A. Point-Prediction Performance

Table III presents the currently reported point-prediction results for the neural network and simple statistical baselines.

The MAE and MSE values in the table are transcribed from the current local evaluation outputs. RMSE values are calculated as the square root of the reported MSE and are rounded to two decimal places.

In the current experiment, the uncertainty-aware neural network has a lower reported MAE than the simple statistical baselines. Its reported MAE is approximately 24.045, compared with approximately 85.314 for the simple mean and median baselines and approximately 85.538 for the simple-average baseline.

The neural network also has a lower reported MSE. These observations indicate that, under the current evaluation setup, the neural network produces smaller point-prediction errors than the listed simple baselines.

However, the comparison should be treated as preliminary. The model outputs, target scaling, baseline construction, and test-set evaluation procedure must be checked to ensure that the comparisons are valid and consistent. The available results alone do not establish that the model will outperform stronger machine-learning baselines or reproduce the published performance.

B. Uncertainty-Related Results

Table IV summarizes the uncertainty- related metrics currently recorded for the evaluated models.

The uncertainty-aware neural network reports a coverage value of approximately 0.8911 and a mean prediction- interval width of approximately 100.199. The simple mean and median baselines report slightly higher coverage, approximately 0.9019, but their reported mean interval width is approximately 350.809.

This comparison illustrates why coverage should not be considered in isolation. A model can produce wider intervals and thereby cover more observations without necessarily providing more informative uncertainty estimates. The neural network's narrower reported intervals are potentially useful, but their quality cannot be established from interval width alone.

The nominal coverage level required for evaluating calibration must be stated explicitly. The interval construction procedure and the implementation of `mean_unc` should also be verified before concluding that the network is well calibrated.

C. Additional Likelihood-Related Measures

Table V reports the additional likelihood-related values currently available in the execution output.

The neural-network output reports higher values for these three measures than the listed baselines. Their interpretation depends on the exact metric definitions implemented in the repository. Accordingly, the values are reported descriptively rather than treated as probabilities or formal evidence of calibration.

D. Correlation Statistics

The first recorded neural-network evaluation includes a Spearman correlation of approximately 0.1127 and a Pearson correlation of approximately 0.2851.

The local output also contains correlation warnings for some baseline and subsequent evaluation outputs. When a correlation calculation receives constant input, the correlation coefficient is undefined. Such results should be reported as undefined or not available rather than interpreted as zero correlation.

TABLE III
REPORTED POINT-PREDICTION RESULTS

Model	MAE	MSE	RMSE
Uncertainty-aware neural network	24.045	1001.015	31.64
Simple median	85.314	13518.207	116.27
Simple mean	85.314	13518.207	116.27
Simple average	85.538	12734.100	112.85

TABLE IV
REPORTED UNCERTAINTY-RELATED METRICS

Model	Coverage	Mean Uncertainty	Mean PI Width	Likely_30
Uncertainty-aware neural network	0.8911	64.934	100.199	0.6549
Simple median	0.9019	60.000	350.809	0.1956
Simple mean	0.9019	60.000	350.809	0.1956
Simple average	0.8909	113.585	336.259	0.1903

TABLE V
ADDITIONAL RECORDED MEASURES

Model	Likely_15	Likely_30	Likely_45
Neural network	0.3679	0.6549	0.8341
Simple median	0.0986	0.1956	0.2896
Simple mean	0.0986	0.1956	0.2896
Simple average	0.0958	0.1903	0.2821

The correlation values should be rechecked using the actual predictions and ground-truth targets from the same evaluation run. This is particularly important because identical point-prediction errors across different evaluation outputs may indicate that the same saved predictions are being reused.

E. Consistency Checks

The current execution logs contain repeated point- prediction metrics for the neural-network model despite differences in some uncertainty-related values. The simple mean and median baselines also have identical reported results.

These observations do not, by themselves, prove that the implementation is incorrect. They do, however, motivate a consistency check of the inference and evaluation pipeline.

The following checks are required:

- Confirm whether the outputs represent different stochastic inference passes, separate model training runs, or repeated evaluation of the same saved predictions.
- Verify whether point predictions are expected to remain fixed while uncertainty estimates vary.
- Confirm whether the simple mean and median baselines are intentionally identical under the implemented target distribution or whether the same predictions are inadvertently reused.
- Recalculate MAE, MSE, and RMSE directly from the stored test targets and predictions.
- Confirm that all models are evaluated on the same test observations and target scale.

Until these checks are complete, the current results should not be described as averages across repeated training runs.

V. DISCUSSION

A. Interpretation of the Results

The current implementation reports substantially lower point-prediction errors for the uncertainty-aware neural network than for the three simple statistical baselines. This is an encouraging preliminary observation, but it should be interpreted within the scope of the available experiment.

A comparison against simple baselines establishes a useful first reference point. It does not establish that the implementation reproduces every result in the original study, nor does it demonstrate superiority over more sophisticated regression and time-series models.

A more complete evaluation should include multiple independent training seeds, suitable non-neural regression baselines, and a documented procedure for selecting hyperparameters. Results should be summarized using means and standard deviations across independent runs where appropriate.

B. Importance of Uncertainty Evaluation

The reported coverage and interval-width values demonstrate that uncertainty-related measures can add information beyond point-prediction error.

Coverage describes how frequently the realized targets fall inside prediction intervals, while interval width describes the size of those intervals. Neither measure alone fully characterizes uncertainty quality.

Future evaluation should specify the intended nominal coverage level and examine calibration at that level. It should also consider proper scoring rules, such as the interval score or continuous ranked probability score, where compatible with the predictive output.

The uncertainty estimates should further be examined against realized prediction errors. For example, observations can be grouped by predicted uncertainty to determine whether higher uncertainty is associated with larger errors. Such an

analysis would provide additional evidence about whether the estimated uncertainty is informative.

C. Reproducibility Considerations

Reproducibility requires more than successfully executing a public code repository. The following information should be documented for the final paper:

1) Dataset source, version, and provenance.
2) Feature selection and preprocessing procedures.
3) Train, validation, and test split methodology.
4) Model architecture and inference settings.
5) Hyperparameter-selection procedure.
6) Definitions and implementations of all metrics.
7) Random seeds and the number of independent runs.
8) Software and library versions.

The original dataset and the local dataset must not be assumed equivalent without evidence. If the local data are a substitute or a modified version, the paper should identify them accurately and describe the work as an implementation or independent evaluation rather than an exact reproduction.

D. Limitations

The current study has several limitations.

First, the provenance and equivalence of the local dataset relative to the original publication require verification. Differences in data, preprocessing, or splitting can materially affect prediction results.

Second, the available results do not yet establish performance across multiple independent training runs. Repeated evaluation of the same saved predictions is not equivalent to repeated training with different random seeds.

Third, the exact definitions of several uncertainty- related metrics have not yet been independently validated against the implementation. Consequently, these values are reported descriptively.

Fourth, the baseline comparison is limited to simple statistical models. Additional baselines are necessary for a stronger assessment of predictive performance.

Finally, this paper does not demonstrate that visual railway defects cause or predict operational train delays. The vision-based literature discussed earlier provides context for a broader research direction, but linking image-derived anomaly events to delay records requires separate data and experiments.

E. Future Work

Future work should begin with a reproducible evaluation protocol that fixes the dataset version, preprocessing, split strategy, and metric definitions. The experiment should then be repeated across independent training seeds, with all runs recorded in a results table.

The next stage should examine uncertainty quality using calibration analysis and error-versus-uncertainty comparisons. Additional regression baselines should be included to determine whether the observed performance gain remains under more demanding comparisons.

A separate extension could investigate the relationship between visual railway anomalies and operational delay prediction. Such an extension would require a reliable mapping between detected anomalies and corresponding train, location, and time records. Any simulated association would need to be identified explicitly and would not establish real-world predictive validity.

VI. Conclusion

This paper presented an initial empirical implementation and evaluation of an uncertainty-aware neural-network approach for train delay prediction. The study was motivated by the uncertainty-aware framework proposed by Spanninger et al. and used a publicly available implementation as its starting point.

In the current local experiment, the neural network reported an MAE of approximately 24.045 and an MSE of approximately 1001.015. The simple mean and median baselines reported an MAE of approximately 85.314, while the simple-average baseline reported an MAE of approximately 85.538. The experiment also recorded prediction-interval coverage, interval width, and additional likelihood-related measures.

These findings suggest that the evaluated neural network performs better than the listed simple baselines under the current experimental setup. Nevertheless, the results remain preliminary. Dataset equivalence, metric definitions, prediction consistency, and repeated-run behaviour must be checked before claiming exact reproduction or generalizable performance.

The study provides an initial basis for a more rigorous evaluation of train-delay prediction models. Future work should prioritize independent repeated runs, stronger baselines, explicit uncertainty calibration, and transparent reporting of the complete experimental protocol.

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